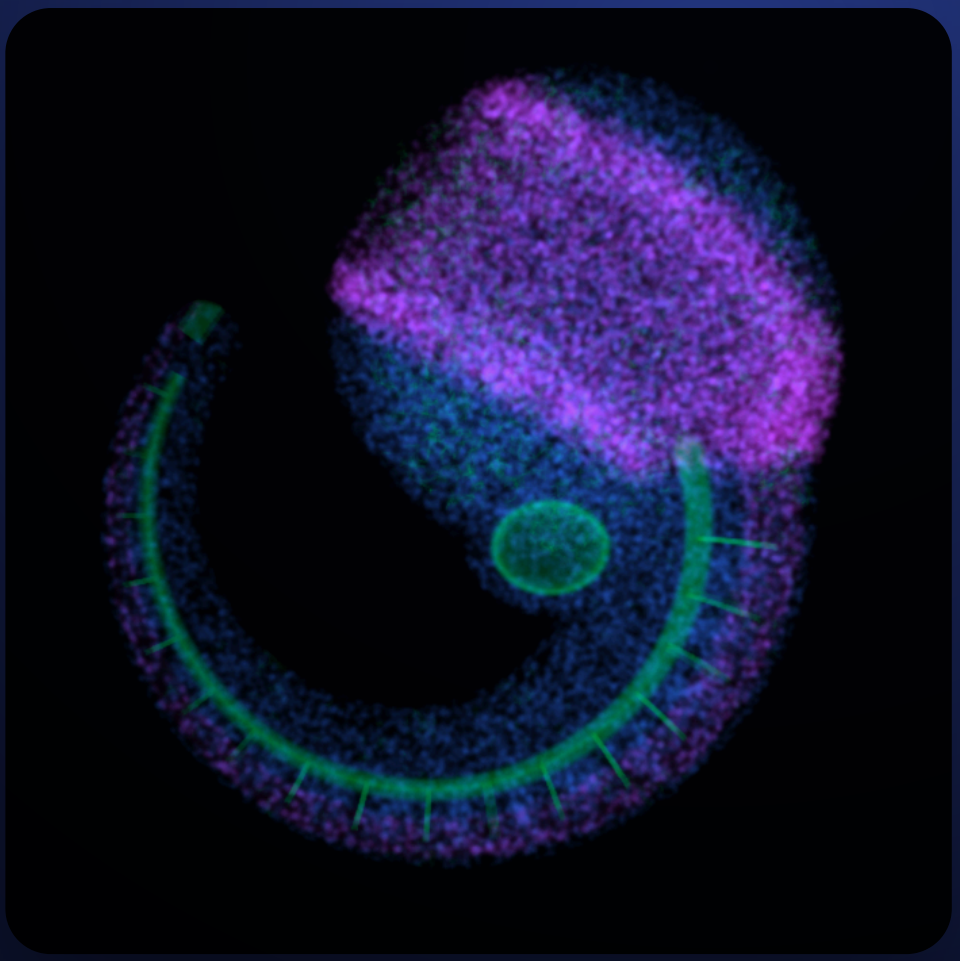


IRIBHM · ULB — LUMEN3D

Lumen3D — the essentials

What happens to your Imaris file, from the microscope to the screen — and how to make good use of it

The biologist’s guide: prepare an acquisition, have it processed, put it online, explore it in 3D, measure, build figures, and know how to interpret what you see. Each section points to the chapter of the full documentation (“Lumen3D, from A to Z”) that gives every detail.

Web platform 1.76.0 · Pipeline 0.23.0 · Data format 6

November 2026

Contents

1. Lumen3D in two pages

- 1.1 Why a preparation step is essential
- 1.2 Three kinds of datasets

2. Before the pipeline: prepare your acquisition

- 2.1 The file name: stage and embryo
- 2.2 What Imaris must contain
- 2.3 Colours, channels, 8- or 16-bit data
- 2.4 Timelapse with cell tracking

3. Running the pipeline

4. Cleaning the image

- 4.1 The five landmarks, one sentence each
- 4.2 A complete example, voxel by voxel
- 4.3 On a real slice
- 4.4 The limit to know: minimum size
- 4.5 For a timelapse: one shared window

5. Reducing the size without loss

- 5.1 The pyramid
- 5.2 Lossless compression
- 5.3 Empty bricks are not stored

6. Bricks and formats

- 6.1 Format 6: lossless for measuring, WebP Q30 for looking
- 6.2 Format 5: a resolution × compression matrix (older datasets)

7. The end of the pipeline

8. Putting the dataset online and setting it up

- 8.1 The settings that matter to you

9. Exploring in 3D

- 9.1 Quality and loading
- 9.2 How the image is computed
- 9.3 The three render modes

10. Adjusting how a channel is displayed

11. Measuring and exploring the inside

12. Making figures and exporting

13. Timelapses and cell tracking

14. 2D photographs

15. Interpreting what you see correctly

- 15.1 What the platform does not do

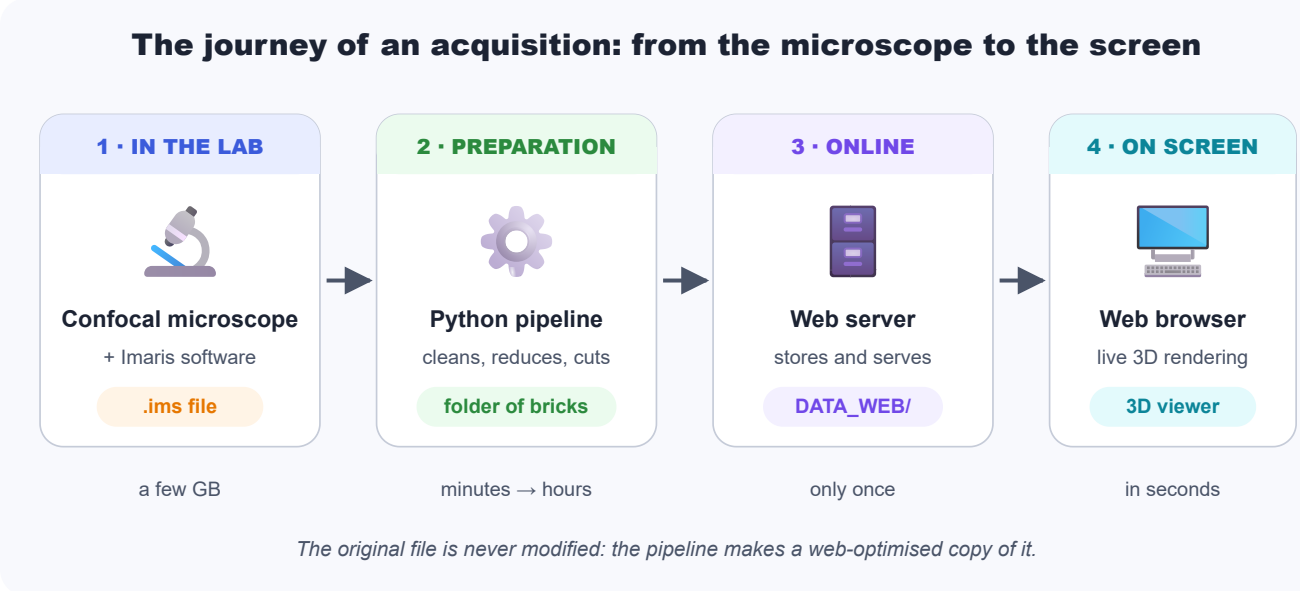
16. What happens if… ?

17. Words to know

1. Lumen3D in two pages

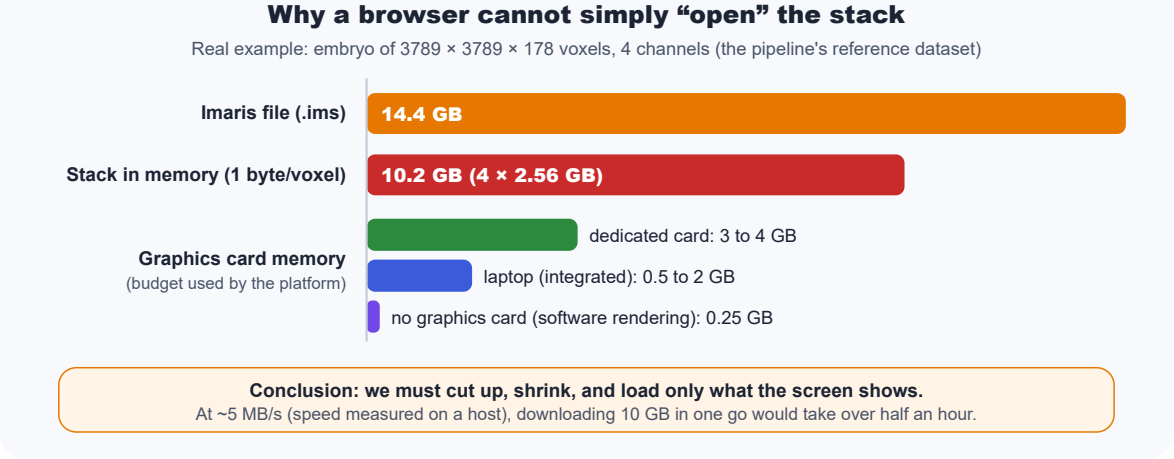
⏱ IN 30 SECONDS

- Lumen3D shows confocal volumes of **several gigabytes** in a plain web browser, with nothing to install.
- A **pipeline** prepares each Imaris file **once**; the original file is never modified.
- The **viewer** downloads only what the screen needs, then computes the 3D image on the graphics card.



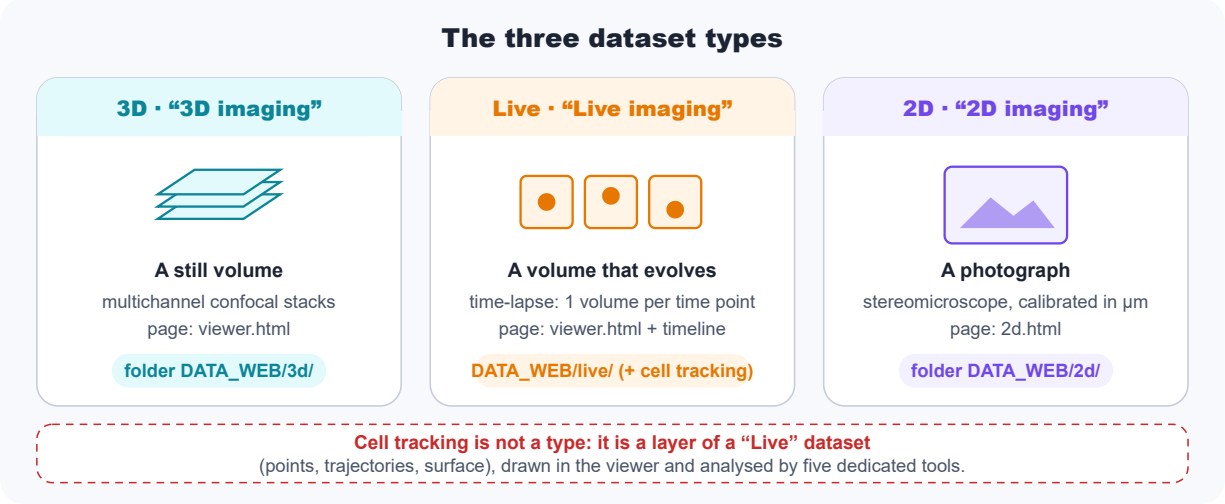
▲ From the microscope to the screen, in four steps.

1.1 Why a preparation step is essential



▲ Even reduced to 1 byte per voxel, a large embryo exceeds the memory of an ordinary computer: it has to be cut up, and only the useful part loaded.

1.2 Three kinds of datasets



▲ Three types, three display pages. The displayed names (3D, 2D, Live) can be renamed by the administrator.

TYPE	WHAT IT IS	PAGE	SPECIAL FEATURE
3D	a confocal stack, up to 4 channels displayed	3D viewer	measuring tools, slices, Studio
Live	a time series (timelapse)	3D viewer + timeline	can carry a **cell tracking**
2D	a calibrated photograph (stereomicroscope)	2D page	X-gal staining isolation

ℹ️ **GOOD TO KNOW**

This document summarises the **full documentation** (“Lumen3D, from A to Z”, 21 chapters). The pointers **→ ch. N** lead to the chapter that explains everything in detail. The volume images come from **synthetic demonstration embryos**, processed by the real pipeline.

2. Before the pipeline: prepare your acquisition

⏱ **IN 30 SECONDS**

- The **file name** gives the stage and the embryo number: choose it carefully.
- The **calibration** (voxel size) and the **channel names** are read from the Imaris file: check them before exporting.
- For a tracked timelapse, prepare the **tracking source** (one of the three accepted formats).

2.1 The file name: stage and embryo

The pipeline reads the stage (`E…`) and the embryo (`Em…`) from the name. The exact rules:

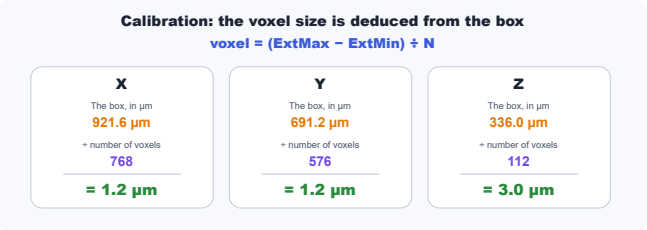
IN THE NAME	STAGE READ	EMBRYO READ
`…-E8.5-Em1-…` **or** `…-E8,5-…`	E8.5	Em1
`…-E8-5-…` **(dash: 25, 5 or 75 only)**	E8.5	—
`…-E85-…` · `…-E105-…` · `…-E95-…` **(compact form)**	E8.5 · E10.5 · E9.5	—
`…-E8-1-DAPI…` **(the “1” is not a fraction)**	E8	Em1
`…-E95-2-…`	E9.5	Em2
no `E…` **recognised**	“Unknown”	—

The published folder name is the file name, cleaned (characters other than letters, digits, `.` , `_` , `-` replaced by `-`). All of this can be corrected later in the administration. → ch. 8

2.2 What Imaris must contain

Check in Imaris before exporting:

- **the calibration** (voxel size in X, Y, Z): without it, no scale bar and no measurement in µm;
- **the name of each channel** (DAPI, Pecam1…): an empty name becomes “Channel 1”, “Channel 2”…;
- for a timelapse, **the time** of each frame: the interval kept is the median of the gaps.



▲ The voxel size is computed: extent ÷ number of voxels, axis by axis. Here 921.6 µm ÷ 768 = 1.2 µm.

🔢 **WORKED EXAMPLE**

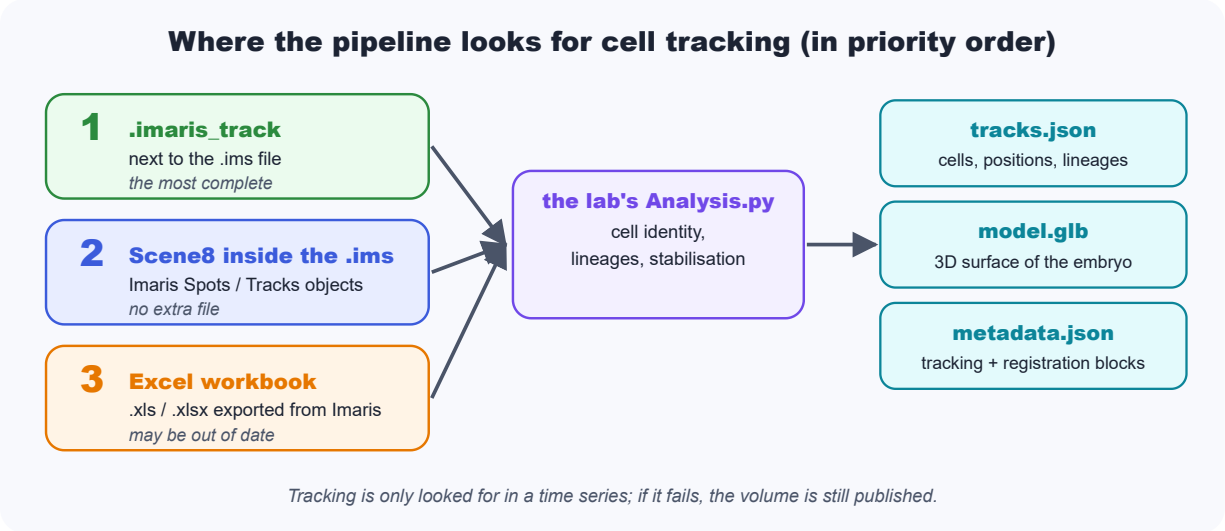
The pipeline reads the **physical extent** (ExtMin → ExtMax) and divides it by the number of voxels. 921.6 µm over 768 voxels → **1.2 µm** per voxel in X. 336 µm over 112 slices → **3.0 µm** in Z. nm and mm are converted to µm. A missing calibration is **never invented**.

2.3 Colours, channels, 8- or 16-bit data

- Imaris **colours** are not carried over: each channel gets a default colour (green, sky blue, magenta, red…), which the administrator then changes.
- **8- or 16-bit** files: both are accepted. The published image is always 8-bit (§ 4).
- More than **4 channels**: all are processed, but the viewer displays only the first 4.

2.4 Timelapse with cell tracking

The pipeline looks for the tracking automatically, **in this order**:

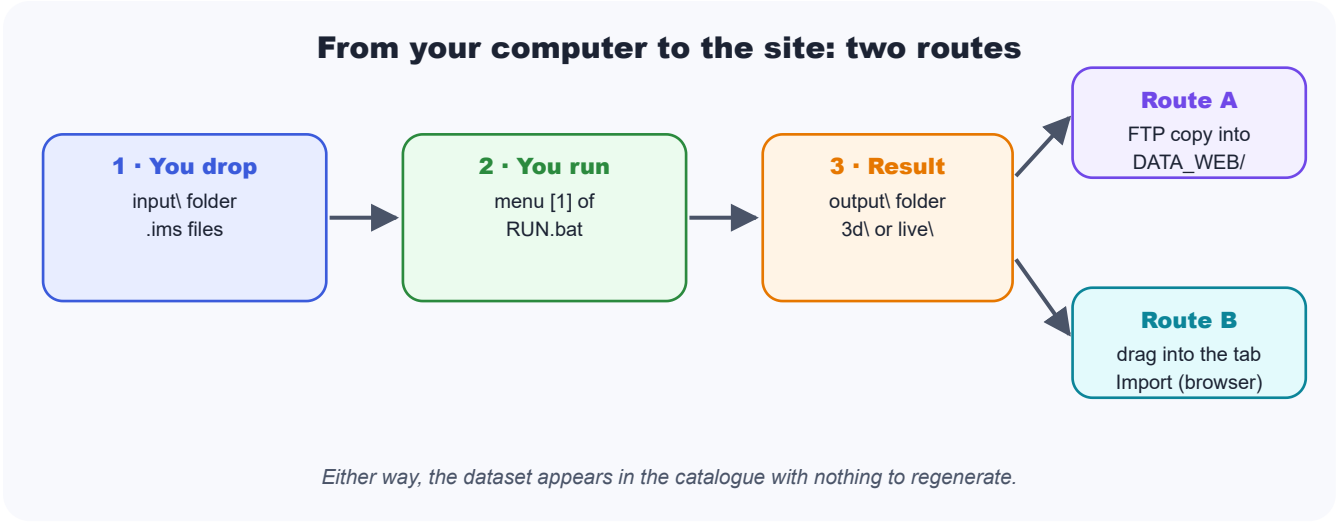


▲ The three possible sources, from the most complete to the most fragile.

> ⚠️ **WARNING**
>
> The Excel workbook exported from Imaris can be **stale** (measured on a lab dataset: 155 of the 172 spots). Its name must contain the interval between two frames (e.g. `30min`): the analysis reads it from the name. Prefer the *Spots/Tracks* objects saved inside the `.ims` itself.

3. Running the pipeline

> ⏱ **IN 30 SECONDS**
>
> - Everything starts from `RUN.bat`, on a Windows PC, from the **Pipeline pack** downloaded in the *Pipeline* tab of the administration.
> - The `.ims` files go into `input\`, the result comes out in `output\`.
> - The dataset is built aside and appears only once **entirely** finished.



▲ From your computer to the site: drop the files, run, then copy or drag.

RUN.BAT MENU CHOICE	WHAT IT DOES
[1]	Process Imaris volumes (`.ims` → `output\` , cell tracking included)
[2]	Imaris tracking analysis (Excel → `tracking\OUTPUT\`)
[3]	Import 2D photographs (`.tif` → `output\2d\`)
[4]	Attach a tracking to an already processed dataset
[5]	Check the environment only

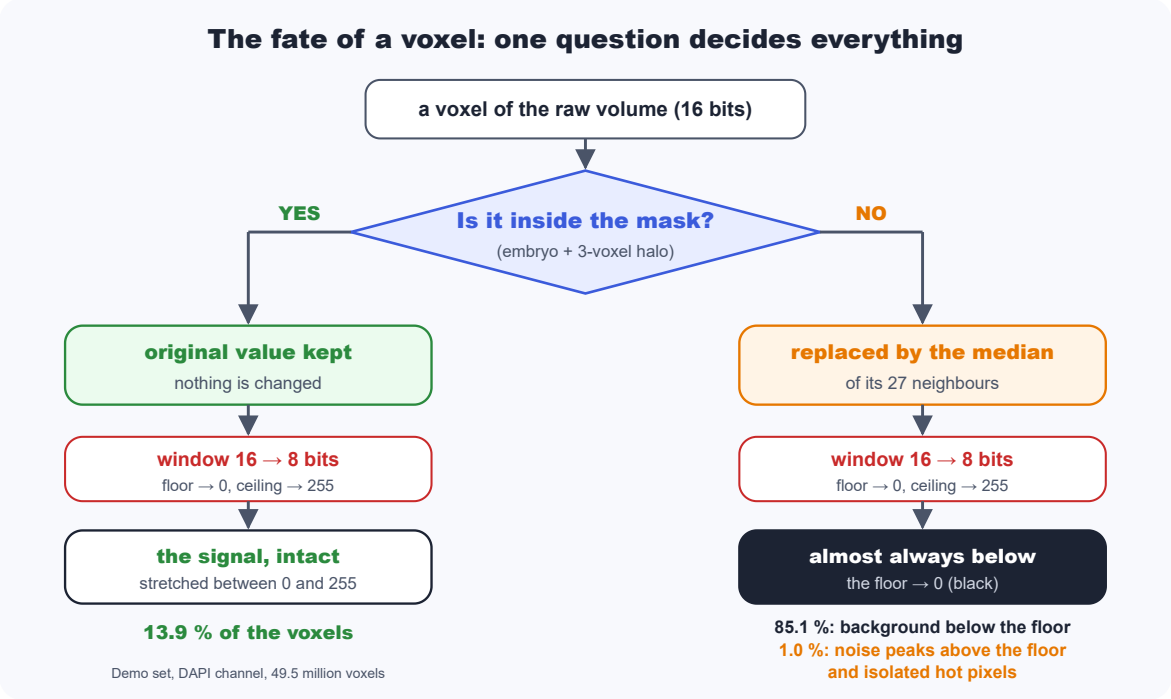
👉 **TIP**

Two editions of the pack: **light** (~3 MB, downloads Python on first run) and **full offline** (~70 MB, everything included). Library versions are pinned: two computers produce the **same bytes** from the same `.ims` . Reprocessing a dataset that is already published **keeps** your settings (name, colours, orientation, default view, gallery, visibility). → ch. 8

4. Cleaning the image

⏱ **IN 30 SECONDS**

- The pipeline asks **a single question** of every voxel: *is it part of the embryo?*
- **Yes** → its value is **not changed**. **No** → it is replaced by the **median** of its neighbours, which almost always brings it to **0**.
- Then everything goes to **8 bits** by a simple rule of three: the background becomes exactly **0**.



▲ The fate of a voxel. Percentages measured over the whole volume of the demonstration dataset (DAPI channel).

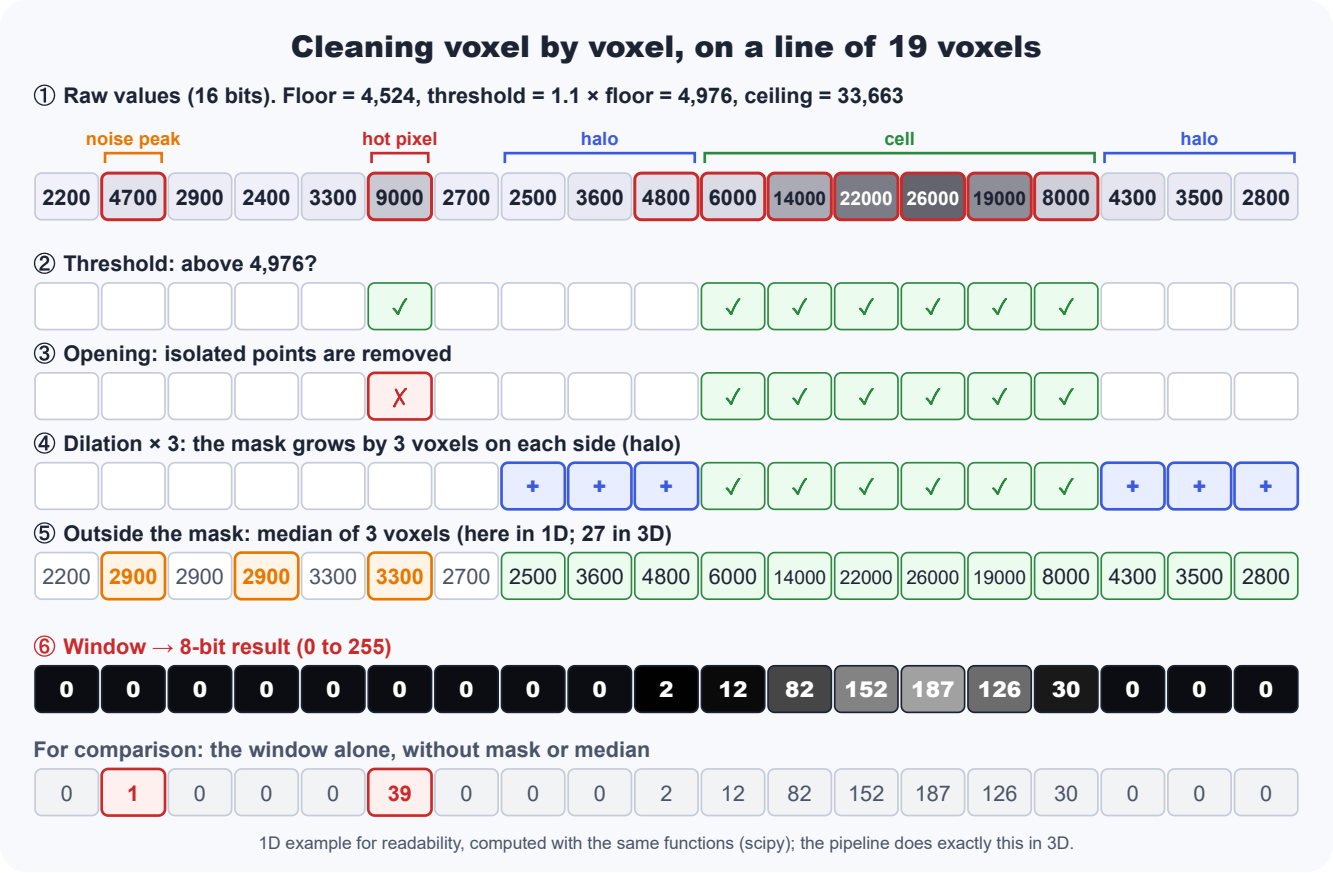
4.1 The five landmarks, one sentence each

	QUESTION	THE PIPELINE'S ANSWER
Floor	Where does signal start?	the value below which **99 %** of the voxels of the **8 corners** of the volume lie (where there is no embryo)
Ceiling	Where should white be?	the value below which **99.9 %** of the voxels lie (1 in 4 in each direction is examined)
Mask	Where is the embryo?	the voxels **above 1.1 × floor**, **without isolated points**, plus a **3-voxel halo**
Median	What to do with the rest?	each voxel outside the mask takes the **middle value** of its 27 neighbours (3 × 3 × 3 cube)
Window	How to go to 8 bits?	floor → **0**, ceiling → **255**, a straight line in between

🔢 **WORKED EXAMPLE**

On the DAPI channel of the demonstration dataset: floor = **4,524**, ceiling = **33,663**. A raw voxel of **20,000** becomes 255 × (20,000 − 4,524) ÷ (33,663 − 4,524) = **135** (decimal part truncated). A voxel at 4,000 becomes **0**; a voxel at 40,000 becomes **255**.

4.2 A complete example, voxel by voxel

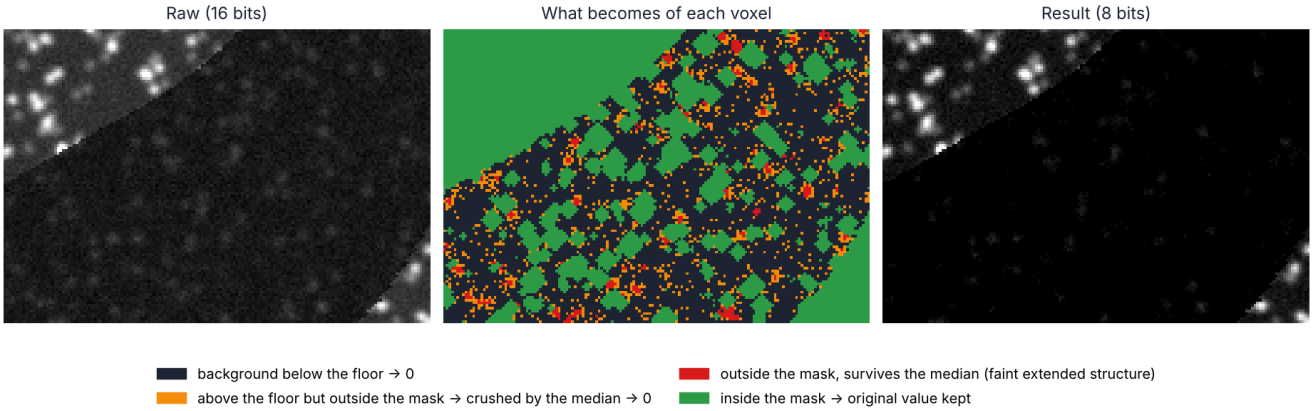


▲ 19 voxels in a row, with the real landmarks of the demonstration dataset. In one dimension so that every number can be read; the pipeline does the same in 3D.

1. **Threshold**: only the cell and the hot pixel (9,000) are above 4,976. The noise peak (4,700) does not pass.
2. **Opening**: the hot pixel is **alone**, it is removed. The cell, wide enough, stays.
3. **Halo**: the mask grows by 3 voxels around the cell; its faint edge (4,800) is protected.
4. **Median** outside the mask: the 4,700 noise peak → 2,900, the 9,000 hot pixel → 3,300.
5. **Window**: everything below the floor → 0. The cell goes from 12 to 187, its faint edge gives 2.

Last row of the figure: **without mask or median**, the noise peak and the hot pixel would stay visible (1 and 39). Over the whole volume, mask + median remove **97 %** of these stray dots (341,674 → 10,069). → ch. 5

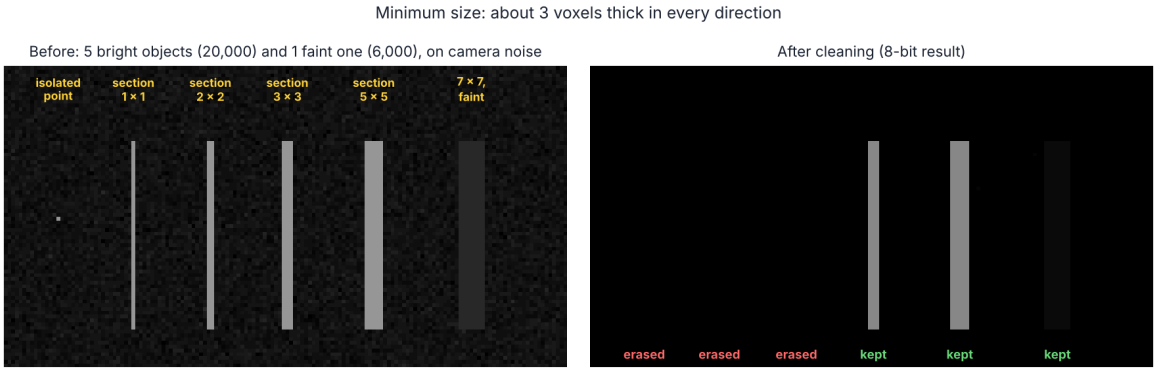
4.3 On a real slice



Slice z = 56, DAPI channel, 160 × 110-voxel area where the tissue is faint (demonstration set).

▲ Left the raw slice, centre what becomes of each voxel, right the result, in an area where the tissue is faint. Green: kept as it is. Orange: crushed by the median. Red (rare): faint but extended area, which survives.

4.4 The limit to know: minimum size

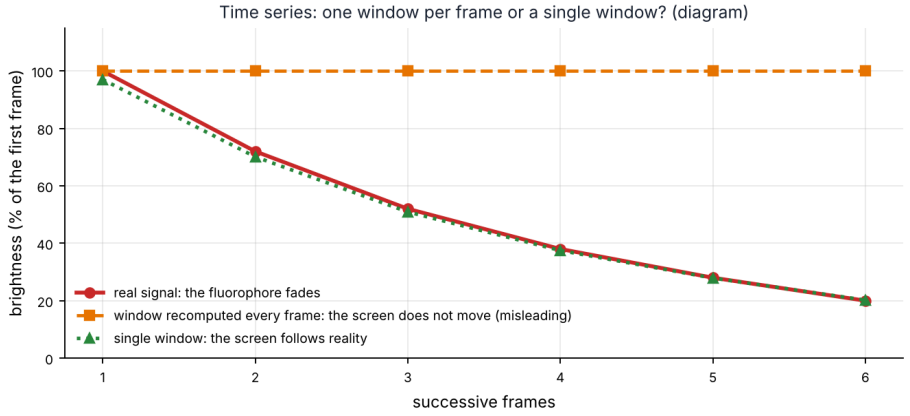


Synthetic test volume, middle slice: each bar has a square cross-section (same thickness in X and in Z). Same floor and ceiling as the demonstration DAPI channel.

▲ Bars with a square cross-section, bright (20,000) or faint (6,000). The 1 × 1 and 2 × 2 cross-sections are erased; from 3 × 3 on, the object is kept, even when faint.

> ⚠ **WARNING**
>
> **An object thinner than about 3 voxels in one direction is erased, however bright** (a point, a thread, a sheet one voxel thick). With 1.2 × 1.2 × 3 µm voxels, that is **3.6 µm in X and Y, 9 µm in Z**: an object present on only one or two slices can disappear. Do the calculation with **your** voxel sizes; when in doubt, compare with the original `.ims`. A thin object **touching** (within 3 voxels) a thicker one is protected by the halo. → ch. 5

4.5 For a timelapse: one shared window



▲ With one window per frame, a fading series would seem to keep the same intensity. Lumen3D uses a single window for the whole series (diagram, illustrative numbers).

The floor and ceiling are computed over **up to 8 frames** spread across the series. The signal level of each frame is **recorded** in `metadata.json`, but no viewer setting uses it yet: a fading series is really displayed darker.

? WHY THIS CHOICE?

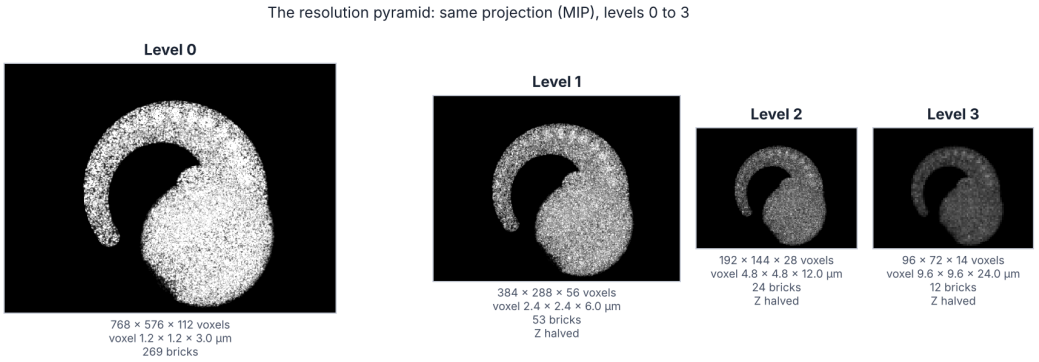
An automatic thresholding method (Otsu) and a neural-network denoiser (Noise2Void) were tried and then **removed** in version 0.12.0: they created artificial coloured “blobs”. → ch. 5

5. Reducing the size without loss

⏱ IN 30 SECONDS

- **8 bits** instead of 16: half as many bytes.
- A **pyramid** of smaller and smaller versions shows something fast, then refines it.
- The volume is cut into **bricks** of 64^3 voxels compressed **losslessly**; empty bricks are **not stored**.

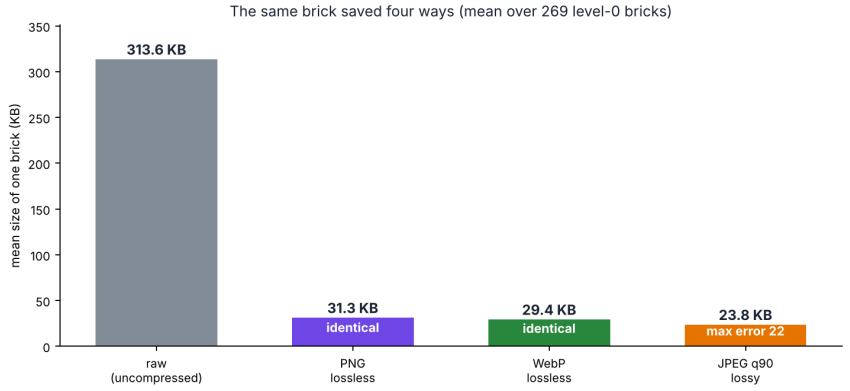
5.1 The pyramid



▲ The four levels of the demonstration dataset, to scale. Each level is the average of 2 × 2 blocks (× 2 in Z when the voxels are fine enough in Z) of the level below.

- In format 4, X and Y are halved at each level; **Z only as long as** the Z voxel does not become more than 1.5 times larger than in XY.
- The average is an **integer, rounded to the nearest**. The pyramid stops when the longest side is 128 voxels or less.
- Level 0 is the cleaned image as it is, **never resampled**. → ch. 6
- This is the rule of **format 4**. **Formats 5 and 6** (the pipeline writes **6**) keep every slice and use sizes of 256 · 2^k voxels: see § 6.1 and § 6.2.

5.2 Lossless compression



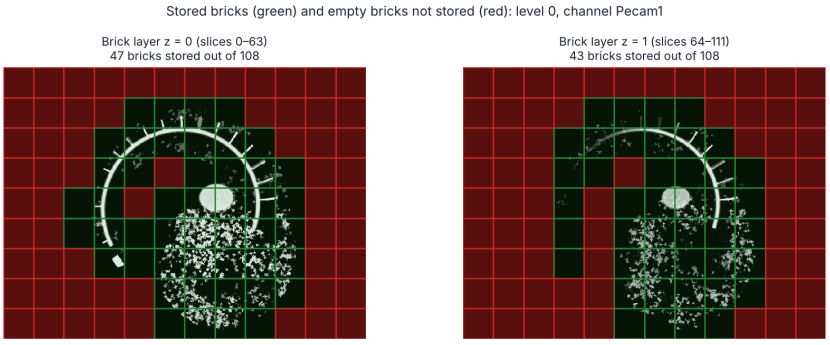
▲ Average size of one brick of the demonstration dataset. Only JPEG is smaller… but it changes the values.

FORMAT	AVERAGE SIZE	EXACT VALUES?
Raw	313.6 KB	yes
PNG (best compression)	31.3 KB	yes
Lossless WebP (chosen)	**29.4 KB**	yes
JPEG quality 90	23.8 KB	**no**: up to 22 levels off, 11.7 % of voxels changed

❓ **WHY THIS CHOICE?**

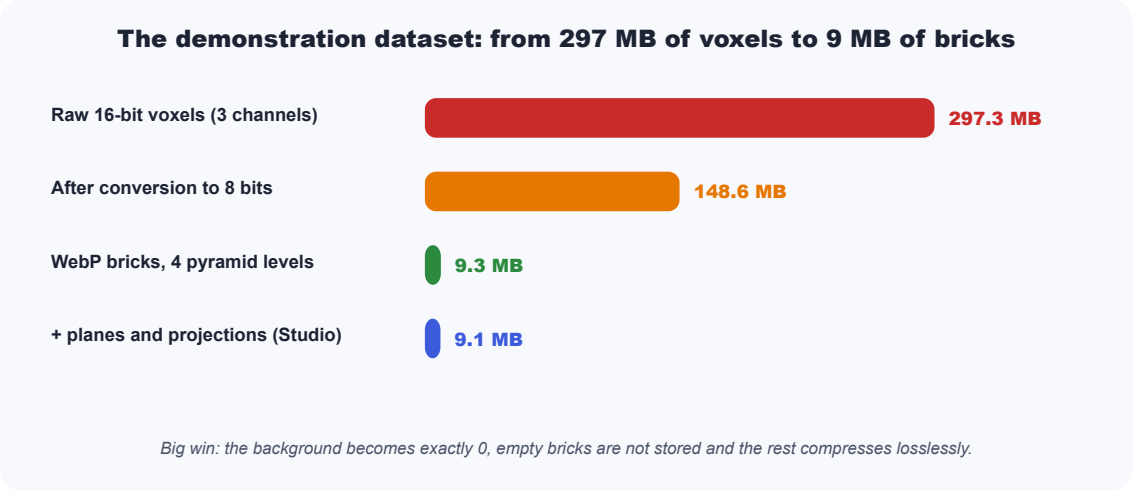
These are **measurements**: lossy compression would invent signal (2.2 % of the voxels at 0 become non-zero in JPEG). WebP is a little more compact than PNG and the browser decodes it very fast. → ch. 6

5.3 Empty bricks are not stored



▲ Green: stored brick. Red: brick whose voxels are all 0, not stored. Demonstration dataset, level 0.

A brick is stored only if **at least one voxel** of its interior is 1 or more. On the demonstration dataset: **358 bricks stored out of 777** possible.

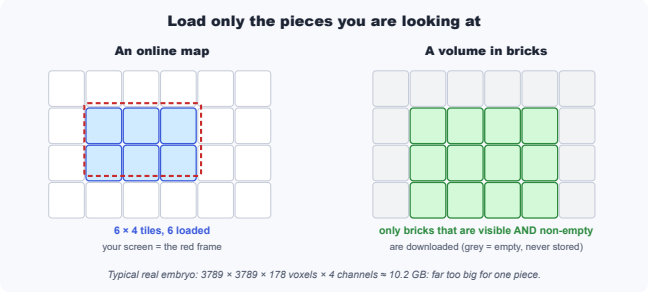


▲ From the original file to the published folder: 297 MB of raw voxels become 9 MB of bricks (demonstration dataset).

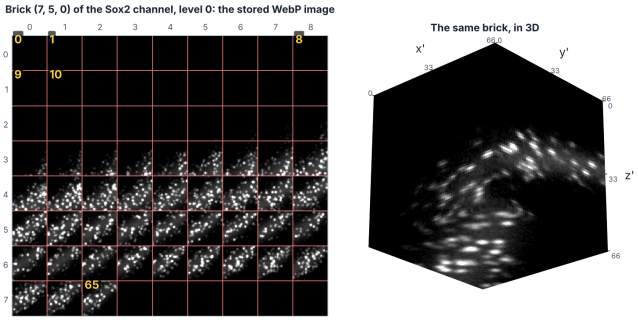
6. Bricks and formats

⏱ **IN 30 SECONDS**

- A brick = **64³ useful voxels** + a **one-voxel border**, laid out flat in **one** WebP image.
- Bricks are grouped into **packs**; a small **index** says where to find each one.
- XY **planes** and **per-layer projections** speed up the Studio’s figures.



▲ Like the tiles of an online map: only the visible, non-empty bricks are downloaded.



▲ A real brick of the demonstration dataset: its 66 slices arranged 9 × 8 in a single image.

- **The border** copies the neighbouring voxel: the image can be smoothed with no visible seam between two bricks. → ch. 7
- **The planes** (`planes/`): a full-resolution XY slice reads **94 KB instead of 5,078 KB** on the demonstration dataset.
- **The formats**: a dataset carries a format number (1 to 6, the latest is **6**). Pipeline 0.23.0 writes format 6 by default; the *Data updates* tab brings an older dataset to format 6, without reprocessing. → ch. 17

6.1 Format 6: lossless for measuring, WebP Q30 for looking

⏱ **IN 30 SECONDS**

- Every resolution is stored **twice**: **lossless** (for exports, downloads and the Studio's figures) and **WebP quality 30** (for the screen).
- The screen shows the Q30 copy: a mean error of **1.5 grey levels out of 255**, and a file about **3.4 times smaller** (26 MB against 89 MB, E8.5 Em1 at native resolution).
- **Fast loading** (§ 9.1) can switch the screen back to the lossless copies.

COPY	USED FOR	SIZE (E8.5 EM1, NATIVE)
Lossless	exports, downloads, the Studio's figures	89 MB
WebP Q30	what you see on screen, by default	26 MB

- The **fast cut stores** (XY, XZ and YZ planes, oblique cut bricks) stay lossless in every format.
- A format-5 dataset becomes format 6 through the *Data updates* tab, one block of bricks at a time. The coarsest levels come first, so they are usable early.

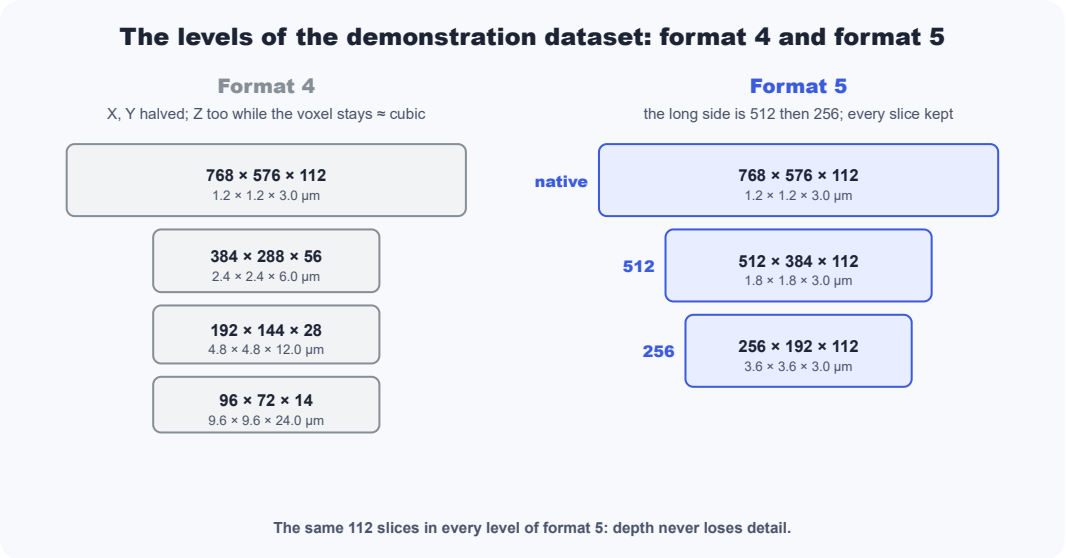
ℹ **GOOD TO KNOW**

Why two copies? A display can be lossy without anyone noticing, but an export cannot. The Q30 copy is what the screen draws, and distances are picked on that picture; the lossless copy is what every exported file is computed from. → ch. 6

6.2 Format 5: a resolution × compression matrix (older datasets)

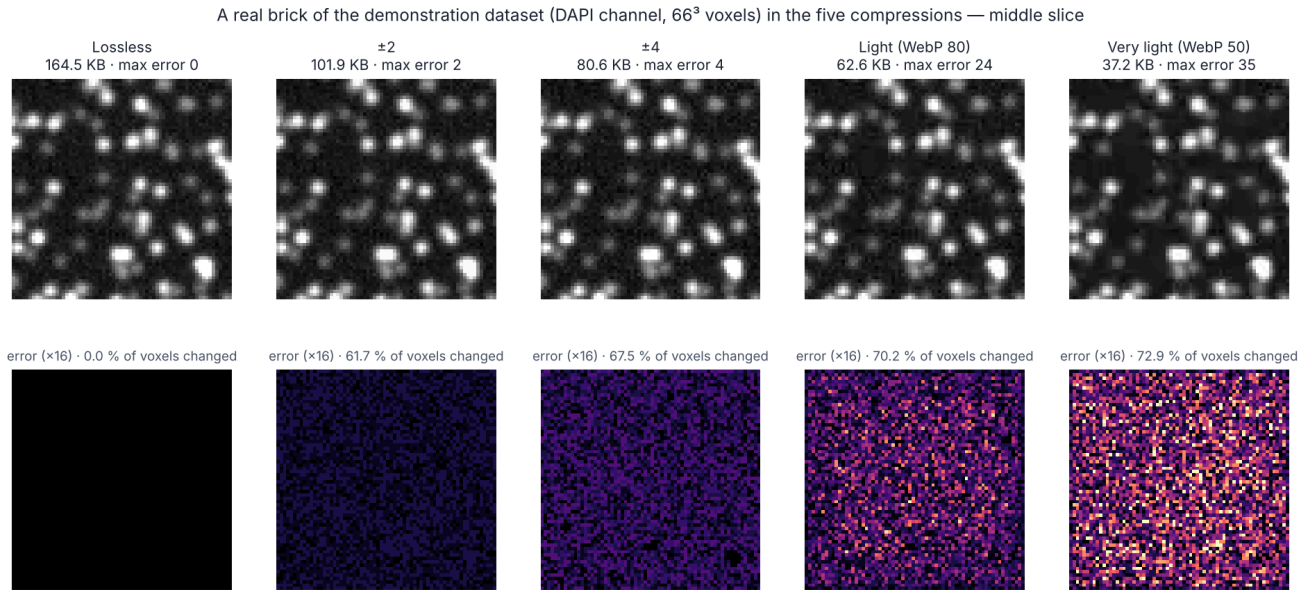
⏱ IN 30 SECONDS

- The same volume is stored at **several resolutions** (the rows) and in **five compressions** (the columns). You pick your cell.
- The reduced levels **keep every slice** (Z is no longer reduced).
- **Distances** are picked on the cell on screen, so they follow it. **Exports** offer a **lossless** version (*Max*), whatever cell is displayed.



▲ The levels of the demonstration dataset in format 4 and in format 5 (real dimensions).

COLUMN	WHAT IS GUARANTEED	SIZE (VS LOSSLESS)
Max (lossless)	every voxel exact	1
Extra (±2)	no voxel moves by more than 2 grey levels (out of 255)	0.62 – 0.82
High (±4)	no voxel moves by more than 4	0.50 – 0.69
Medium (WebP 80)	lossy WebP (quality 80): small visual changes	0.35 – 0.61
Low (WebP 50)	lossy WebP (quality 50): visible smoothing	0.22 – 0.40



▲ The same real brick in the five compressions: top, the middle slice; bottom, the difference from the original (× 16).

- The **background stays exactly black** in every column: ±2 and ±4 keep 0 apart; the lossy columns put back to 0 the small values that “bleed” (a **dead zone**).
- Three **cut copies** (XY planes, XZ / YZ rows, small oblique bricks) make the Studio’s native cuts fast: a native oblique cut read 23 MB instead of 60 on a 1946² × 72 embryo. → ch. 7
- The price: **more disk space** (9 MB of bricks → about 80 MB for the demonstration dataset). What each visitor downloads, on the other hand, goes down.

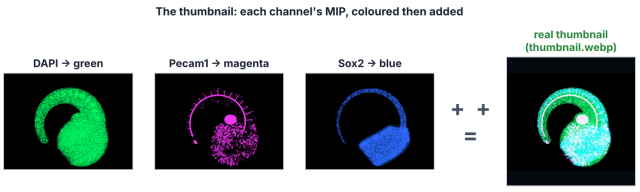
7. The end of the pipeline

⏱ **IN 30 SECONDS**

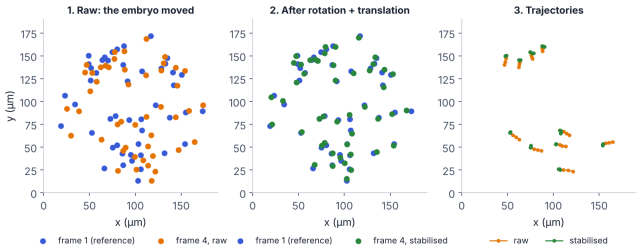
- The pipeline writes a **record** (`metadata.json`), a **thumbnail**, the channel **histograms**.
- It attaches the **cell tracking** of a timelapse and **stabilises** the positions.
- It publishes **all or nothing**: never a half-copied dataset.

PUBLISHED FILE	ROLE
`metadata.json`	the record: name, stage, dimensions, calibration, channels, format
`thumbnail.webp`	the explorer thumbnail (false-colour maximum projection)
`bricks/`	the bricks of every level and their index
`planes/` , `mips/`	XY planes and per-layer projections of 64 planes
`tracks.json` , `model.glb`	cell tracking and embryo surface (timelapses)

PUBLISHED FILE	ROLE
`download/`	(option) the original `.ims`, an ImageJ TIFF, PNG projections, a README



▲ The thumbnail: three maximum projections, coloured and added.



▲ Stabilisation: a rotation + translation (no deformation) superimposes the cells of two time points.

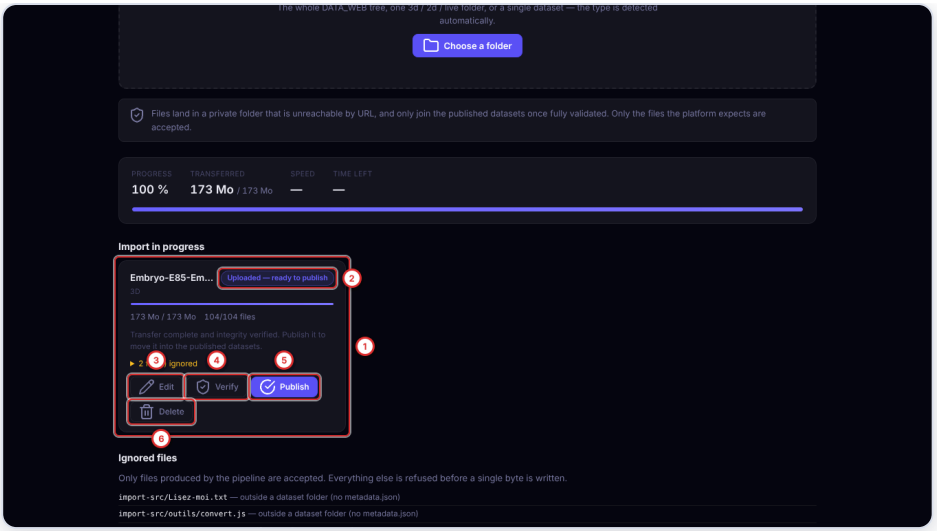
ℹ️ **GOOD TO KNOW**

Stabilisation. The embryo moves during the acquisition. For each time point, the pipeline computes the **rigid** displacement that best superimposes the cells (Kabsch method, at least 3 reference cells). The viewer applies this displacement **to the view**, not to the voxels: no voxel is resampled. → ch. 8 and 13

8. Putting the dataset online and setting it up

⏱ **IN 30 SECONDS**

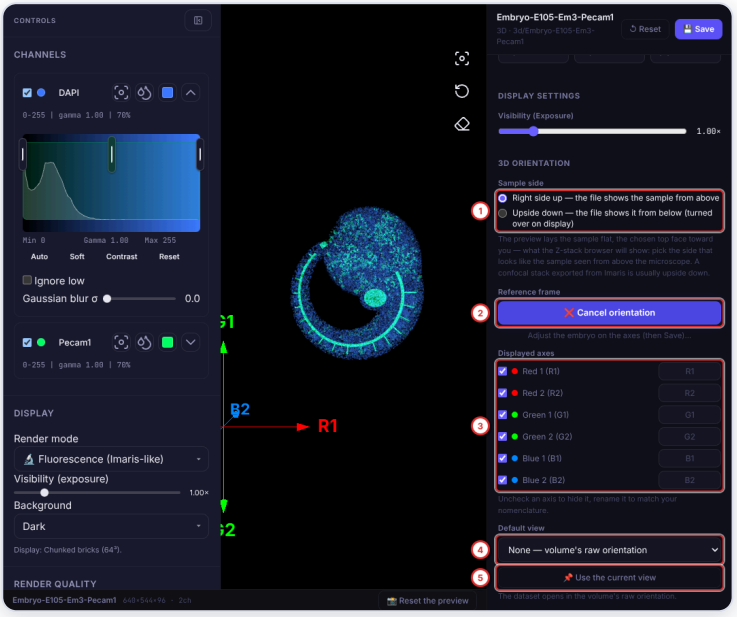
- Two routes: **copy** the folder into `DATA_WEB/` on the server, or **drag** it into the *Import* tab of the administration.
- An imported dataset is **hidden** until you show it.
- The administration sets the name, stage, colours, calibration, **orientation** and **default view**.



▲ The Import tab: a dataset fully transferred and verified, ready to publish (Edit, Verify, Publish, Delete buttons).

- The transfer **resumes by itself** after an interruption: just drag the same folder again (within 7 days).
- The upload order is designed for you: the record, the thumbnail and the **coarsest level** go first, so the dataset can be **set up within minutes**, while the rest arrives.
- Only files produced by the pipeline are accepted. → Administrator guide, ch. 4
- The *Data integrity* tab checks the files of every dataset (missing or extra), and the *Storage* tab shows where the disk space goes. → Administrator guide

8.1 The settings that matter to you



▲ Orientation in the administration: sample side, axis frame, displayed axes, default view.

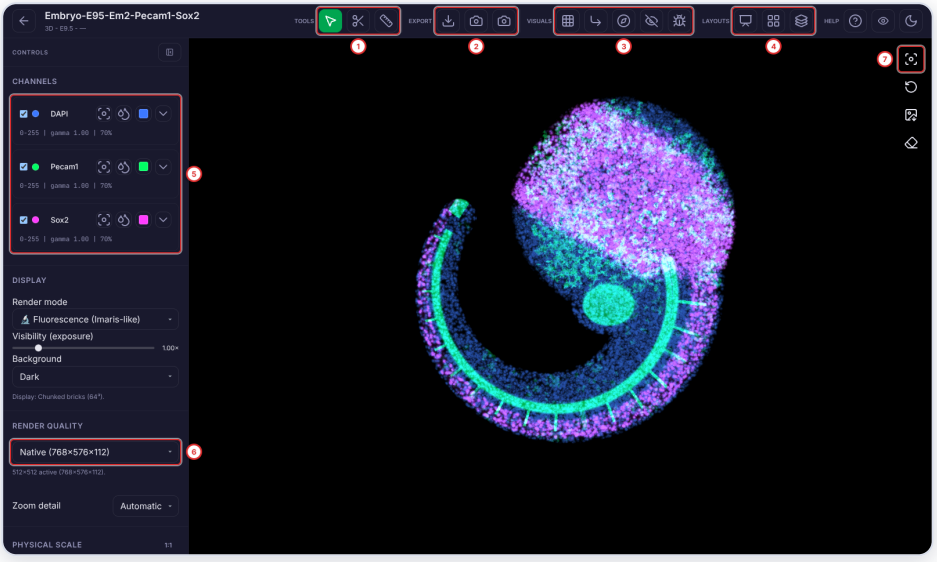
SETTING	WHAT IT IS FOR
Name, stage, embryo, description	what visitors read in the explorer
Channel colours	the default colour of each channel when the dataset opens

SETTING	WHAT IT IS FOR
Calibration	fix a missing or wrong voxel size
Sample side	“Upside down” for a stack seen from below (common for an Imaris export): the Z-stack browser will show the right face
Define orientation	align the red / green / blue axes with the anatomy, then rename them (anterior, dorsal…)
Default view	the pose in which the dataset opens
Gallery	attach annotated figures, visible in the viewer
Visibility	show or hide the dataset in the public explorer

9. Exploring in 3D

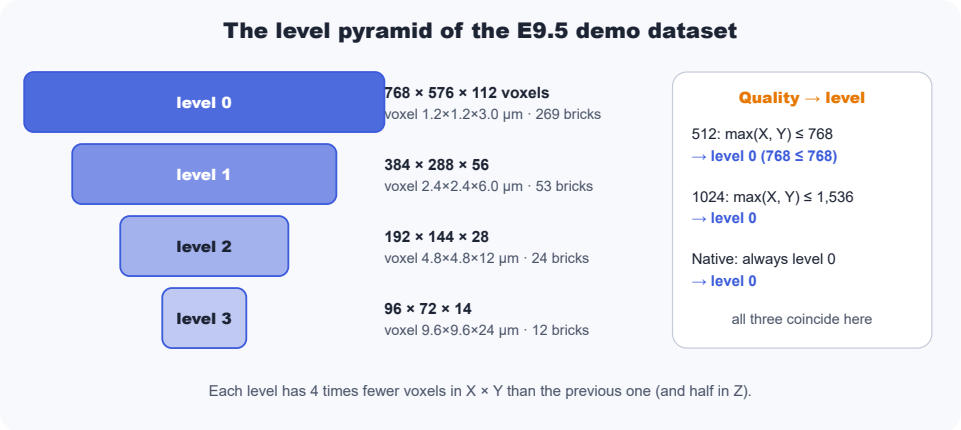
⏱ **IN 30 SECONDS**

- Open a dataset from the **explorer** (search, filters by type and stage).
- The viewer first shows a **coarse image**, then the detail arrives **starting from the centre**.
- Three **render modes** for three different questions.



▲ The viewer: toolbar at the top, channels on the left, the volume in the centre (demonstration dataset).

9.1 Quality and loading



▲ The real pyramid of the demonstration dataset and the rule of the three qualities.

⏱ **IN 30 SECONDS**

- A **format 6** dataset (the current format) lists its **resolutions**. The screen shows the **Q30** copies.
- **Fast loading**, on by default, keeps that Q30 display. Switch it off to see the lossless copies (larger downloads).
- The viewer **detects** the graphics memory of your card. Nothing to set.

DATASET	QUALITY MENU	WHAT THE SCREEN SHOWS
Format 6	the resolution list + *Fast loading*	WebP Q30 (on) or lossless (off)
Format 5	two lists: *Resolution* and *Quality*	the quality you pick
Format 4 (older)	the three presets below	exact voxels

Format 6: resolutions and Fast loading

Each resolution shows its name, its size in voxels, and the graphics memory it takes once loaded.

RESOLUTION (FORMAT 4 PRESETS)	LEVEL LOADED
512 (default)	the finest level whose longest XY side is at most **768** px
1024	the finest level whose longest side is at most **1,536** px
Native	level 0, full resolution

ℹ️ GOOD TO KNOW

Fast loading (the checkbox under the list, on by default, kept in your browser): - **On**: the screen uses the WebP Q30 copies. Smaller downloads, mean error 1.5 grey levels out of 255. - **Off**: the same resolutions load from the lossless copies. Larger downloads, exact voxels on screen.

The volume reloads when you change it. Exports read the lossless data in both positions; distances are picked on the picture on screen, so they follow this switch.

👉 TIP

When to switch Fast loading off. When you compare the screen with the raw values of a faint structure, or when a picture must match a number exactly. For the overview, leave it on.

Format 5: the quality matrix and “How to choose?”

On a format 5 dataset, the menu has two lists: **Resolution** (the rows) and **Quality** (the columns). Each quality shows its download size on this dataset.

QUALITY	MEANING
Max	lossless: every voxel as acquired, for measurements and figures
Extra	±2 grey levels (out of 255): visually lossless, the default for exploring
High	±4 grey levels: near lossless
Medium	WebP quality 80: smaller files, faint texture smoothed
Low	WebP quality 50: the lightest, visible smoothing



▲ The quality matrix of the demonstration dataset (sidebar widened for the capture; on screen, scroll the table to the right).

The **How to choose?** button opens a guide. For each resolution and quality it gives the **download size**, the **download time** at 5, 20 or 100 Mb/s (and at the speed measured in your browser, once measured), and the **graphics memory**.

COLOUR (GPU MEMORY)	MEANING
green	fits comfortably: under 75 % of the budget
amber	close to the budget: 75 to 100 %
red	more than the budget: the viewer shows the next coarser level

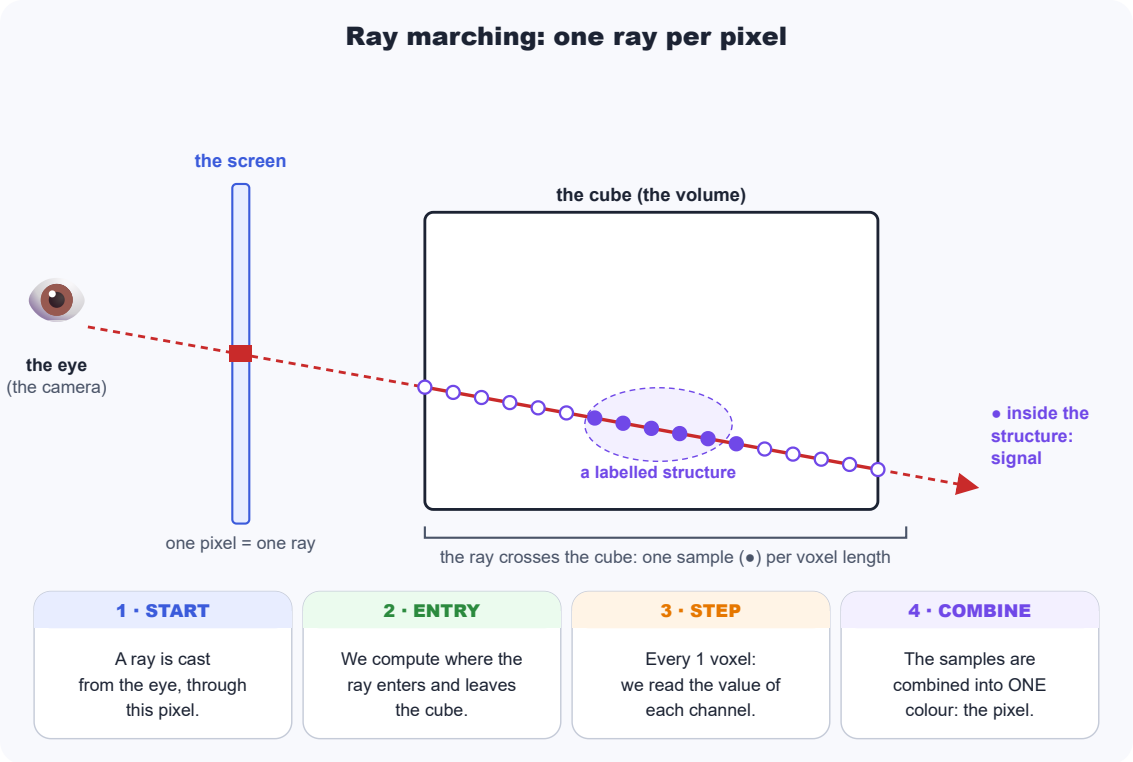
> 👉 **TIP**
>
> **Recommended**: the finest resolution shown in green, at *Extra*. For a measurement or a figure: *Max* at the native resolution.

Graphics memory: detected, not chosen

The browser does not say how much memory the card has. The viewer reads the card’s name, looks its memory up in a table of NVIDIA, AMD and Intel Arc models, and allows **60 %** of it. It always keeps **2 GiB** free for the browser and the system, and never allows more than **16 GiB**. An unknown card falls back to an estimate by class.

If the card cannot hold the level you chose, the viewer shows the next coarser one and a warning says so. A progress line shows the quality, the percentage and the current step. → ch. 10

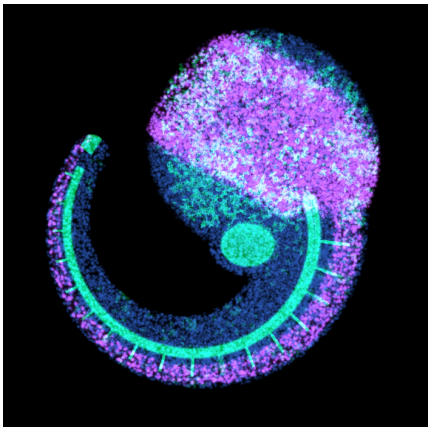
9.2 How the image is computed



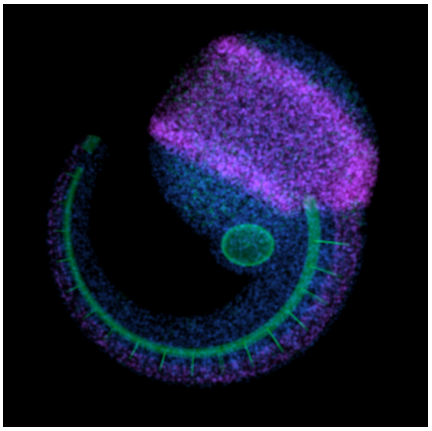
▲ One ray per pixel: it enters the volume, takes one sample per voxel crossed, and combines these values.

The image is not a photo: it is **recomputed** by the graphics card, thousands of pixels in parallel (Three.js + WebGL2). The definition **does not change** while you move: a drag and a still view draw the same picture, at full resolution, with one sample per voxel. Empty space is skipped in cells of 8 × 8 × 8 voxels. → ch. 9

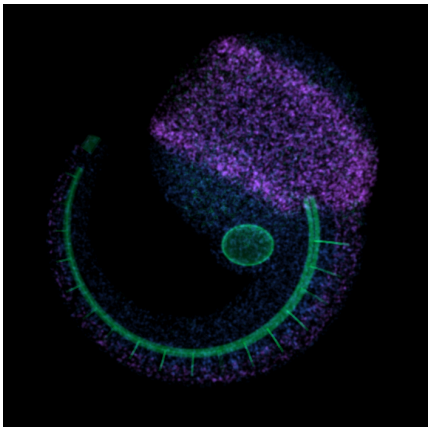
9.3 The three render modes



▲ Fluorescence (Imaris-like), the default.



▲ Natural fluorescence.



▲ Structure (DVR).

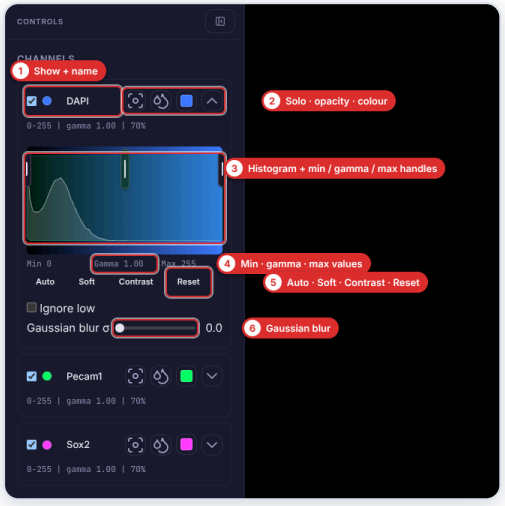
MODE	WHAT YOU SEE	GOOD FOR…
Fluorescence (Imaris-like)	the **maximum** of each channel along the ray, colours added, no depth	seeing all the signal, like a MIP

MODE	WHAT YOU SEE	GOOD FOR…
Natural fluorescence	each fluorophore glows; what is dense **hides** what lies behind	perceiving shape and depth
Structure (DVR)	an opacity accumulated from front to back	surfaces and solid volumes

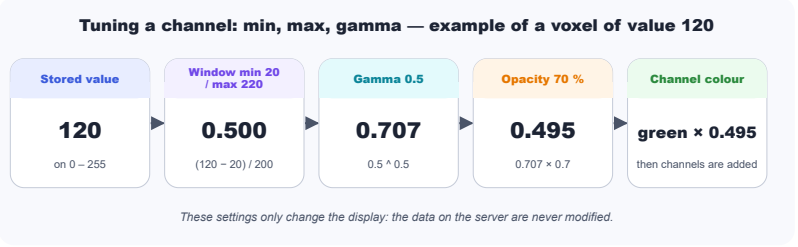
> 👉 **TIP**
>
> **Fluorescence responds at once.** Changing min, max, gamma, opacity or a colour, or switching a channel on or off, redraws the picture without recomputing the volume. The two other modes recompute it at each change.

10. Adjusting how a channel is displayed

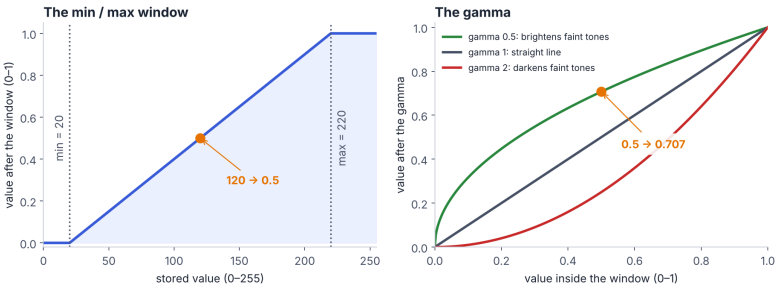
> ⏱ **IN 30 SECONDS**
>
> - Each channel has a **min**, a **max**, a **gamma**, an **opacity** and a **colour**; channels are **added**.
> - The **histogram** shows how the 0–255 values of the coarsest level are spread, in 64 bars.
> - These settings change **only the display**, never the data.



▲ A channel card, unfolded.



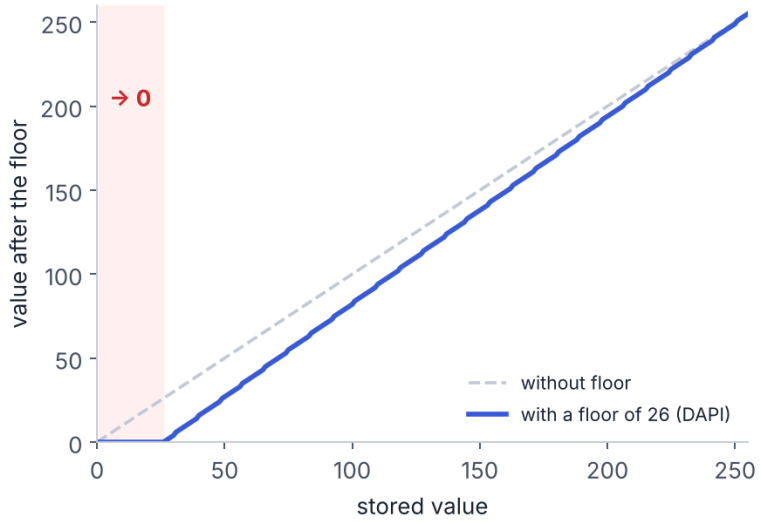
▲ The computation of one voxel, from storage to screen.



▲ Left, the window min 20 / max 220; right, the effect of gamma 0.5 / 1 / 2.

- **Min / Max**: everything below *min* becomes black, above *max* full colour.
- **Middle handle**: the level displayed at 50 %; it sets the gamma (gamma = ln 0.5 ÷ ln *m*).
- **Auto** clips the darkest and brightest 0.5 %; **Soft**, **Contrast** and **Reset** are presets.
- **Opacity**: 70 %, 42 % or 100 %; **Exposure**: from 0.2× to 5× for all channels; **Solo** isolates one channel.
- **Gaussian blur**: plane-by-plane smoothing (σ in voxels), available at qualities that fit in a single texture.
- A **colour-blindness simulation filter** checks that a figure stays readable.

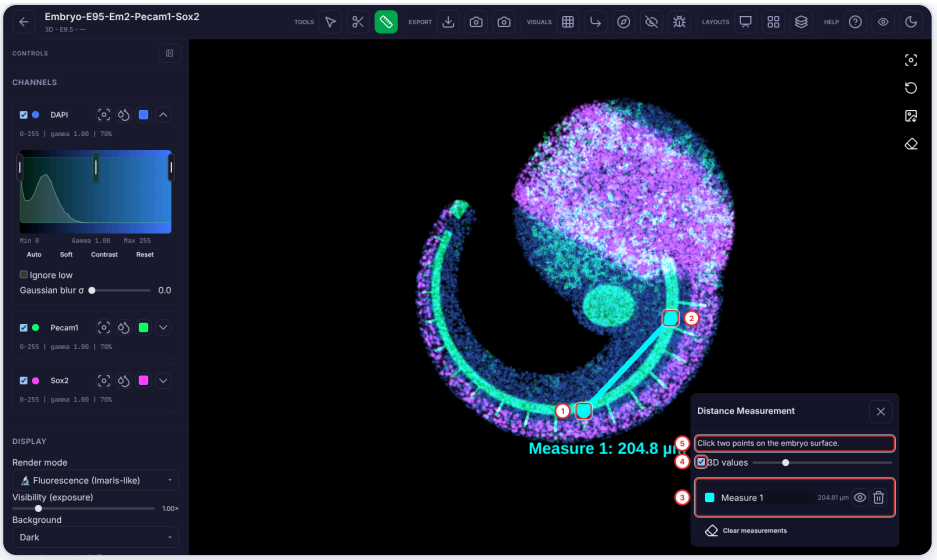
> ⚠ **WARNING**
>
> **A hidden setting.** Before your settings, the viewer also sets the faintest values of each channel to 0 (a “floor” between 6 and 48, estimated from the histogram), then stretches the rest. This is why the residual noise disappears on screen. → ch. 11



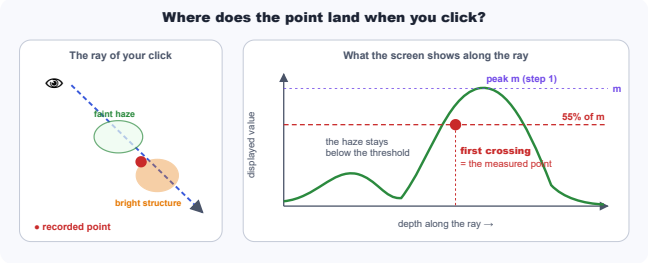
▲ The effect of the display floor: estimated at 26 for the DAPI channel of the demonstration dataset.

11. Measuring and exploring the inside

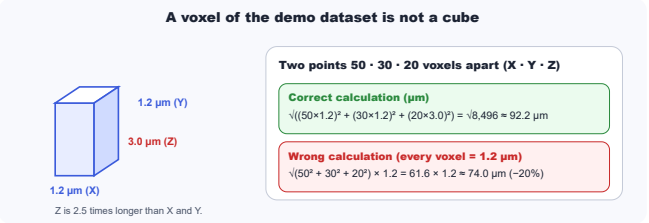
> ⏱ **IN 30 SECONDS**
>
> - **Measurements** are in **µm**, picked on the first visible structure under the click, on the picture on screen; their precision is the voxel size of the level displayed.
> - **Oblique slice** and **Z-stack browser** show the inside, with an exact scale bar.
> - The scale bar of the **3D view** is exact only at the depth of the sample’s centre.



▲ Two points and the measured distance: 204.8 µm (demonstration dataset).



▲ The measured point is the first structure that reaches 55 % of the displayed maximum along the ray. A click in empty space is refused.



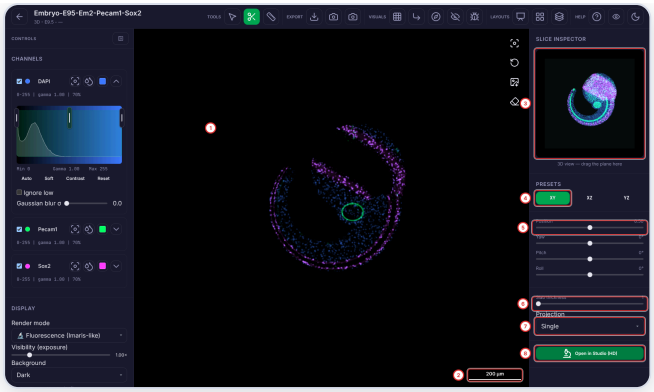
▲ Voxels are not counted: in Z, a voxel is 2.5 times longer here.

🔢 **WORKED EXAMPLE**

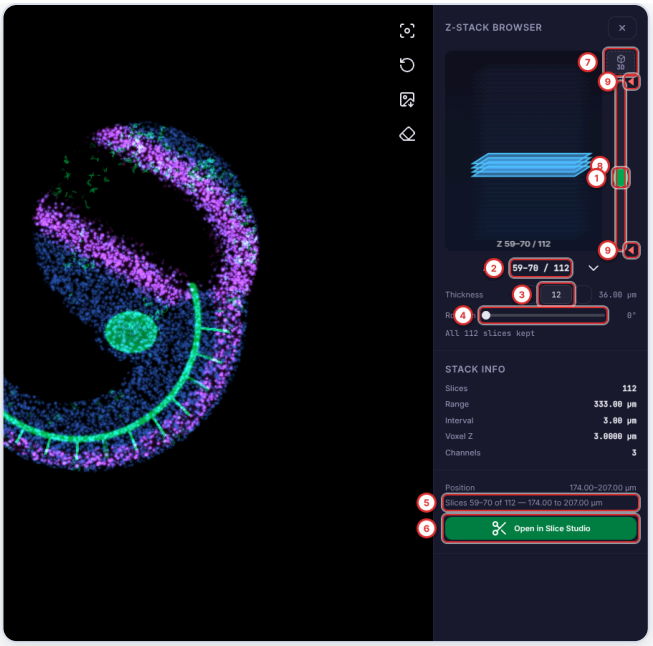
Two points 10 voxels apart in X and 4 voxels apart in Z, with voxels of 1.2 × 1.2 × 3 µm: √((10 × 1.2)² + (4 × 3)²) = √(144 + 144) = **17 µm** — not √(10² + 4²) ≈ 10.8 “voxels”.

⚠️ **WARNING**

Measurements are not saved on the server: closing the tab erases them. Keep them by copying the page address (they travel in the link) or by downloading the **measurements CSV** from the Download Center.

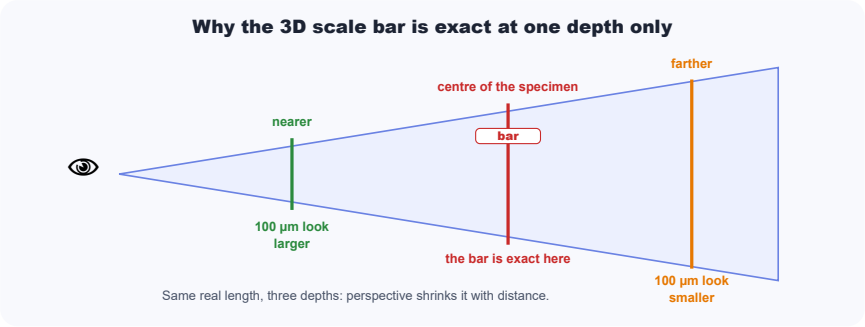


▲ XY slice through the middle of the volume, with its scale bar.



▲ Z-stack browser in slice mode: 12 slices (36 µm).

- **Oblique slice**: a plane in any direction, shown large with an exact scale bar.
- **Z-stack browser**: it lays the stack flat (top face toward you), an adjustable thickness bar, handles to trim the top and bottom, a rotation from 0 to 360°. It is exclusive with the oblique slice.
- **Orientation axes**: the red / green / blue compass (R1/R2, G1/G2, B1/B2) defined by the administrator. They **start at the centre of gravity** of the specimen (the centre of its stored bricks, not of the empty box). Dragging them moves them; the reset action brings them back to the centre.

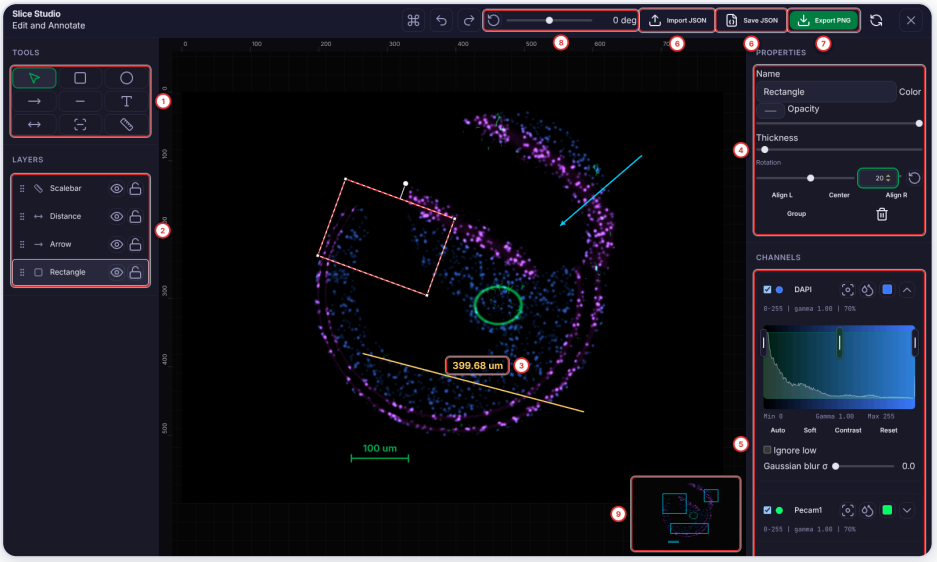


▲ Same length, three depths: the 3D view’s bar is valid for the plane through the sample’s centre.

12. Making figures and exporting

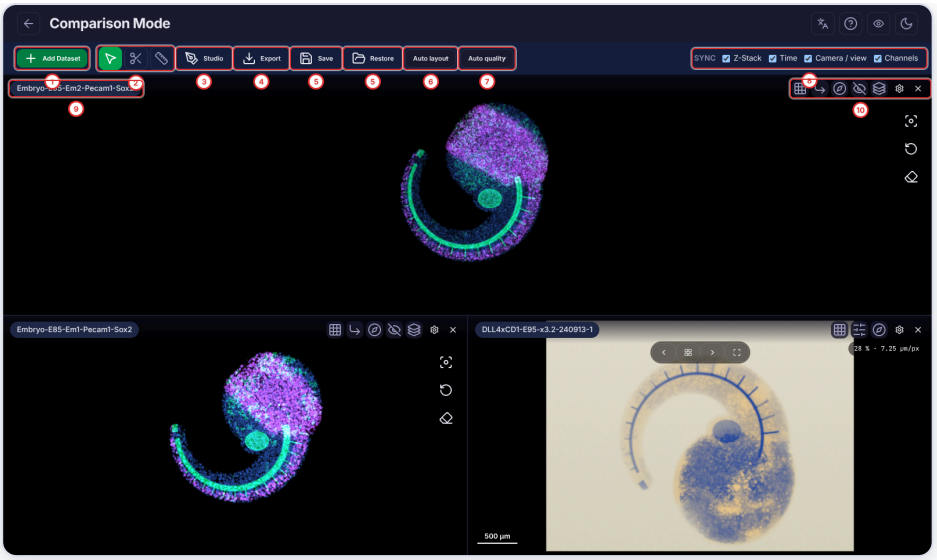
⏱ **IN 30 SECONDS**

- The **Studio** composes an annotated figure at **full resolution** (arrows, text, distances, scale bar).
- On **format 6**, every export is **lossless at the resolution on screen**; the 3D view exports **at any size**, with a transparent background if needed.
- **Compare** puts up to 4 datasets side by side, synchronised.



▲ The Studio: rotated rectangle, arrow, distance, scale bar and text, on a slice of the demonstration dataset.

- On **format 6**, the Studio shows the Q30 picture; the **lossless** native pass runs when you export the figure. On **format 5**, opening a slice reloads it at native level (XY planes or byte ranges), with a progress bar.
- **Channel colours** can be adjusted in the Studio without touching the annotations.
- Export as **PNG** (the figure) or **JSON** (to resume it later); above 5 MB, the export is refused with a message.
- **On format 6, exports are lossless at the resolution on screen:** the 3D view image, the Studio figure and *Decompose by channel* read the lossless copy with no question asked. On **format 5** the 3D view (`Quality` : `Current` or `Lossless`) and the Studio still offer the choice. Your annotations keep their place.
- **Channel Split** (*Decompose by channel*): while the panel is open, a **red frame** on the 3D view shows exactly what the main picture of the figure captures.



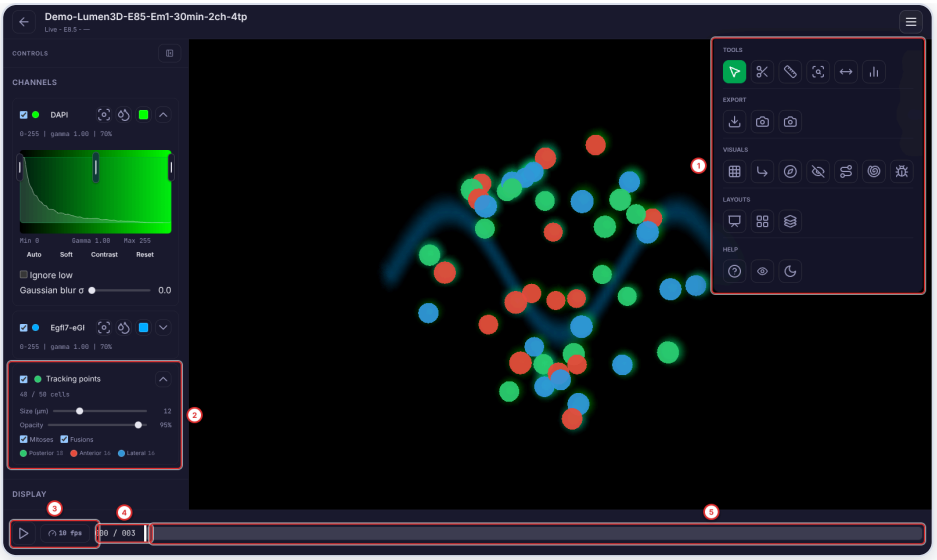
▲ Compare: two volumes and a photograph, shared tools and synchronisation (camera, time, slices, channels).

TO…	USE
get a large, sharp image of the 3D view	**Export view** (rendered in 2,048 px tiles, transparent background possible)
take a quick capture	**Screenshot**
get the original files (`.ims` , TIFF, MIP)	**Download Center** (if the `download/` folder exists)
get the lossless voxels for napari, Fiji or Neuroglancer	**Download Center**: the OME-Zarr export
get the display files as stored (WebP Q30)	**Download Center**: *Download .q30.zip* (format 6 only)
compare embryos or channels	**Compare** and **Decompose by channel**
share exactly what you see	the **page address**, or the workspace saved in Compare

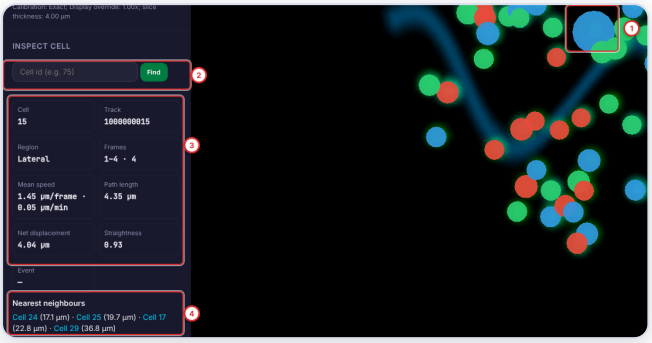
13. Timelapses and cell tracking

⏱ **IN 30 SECONDS**

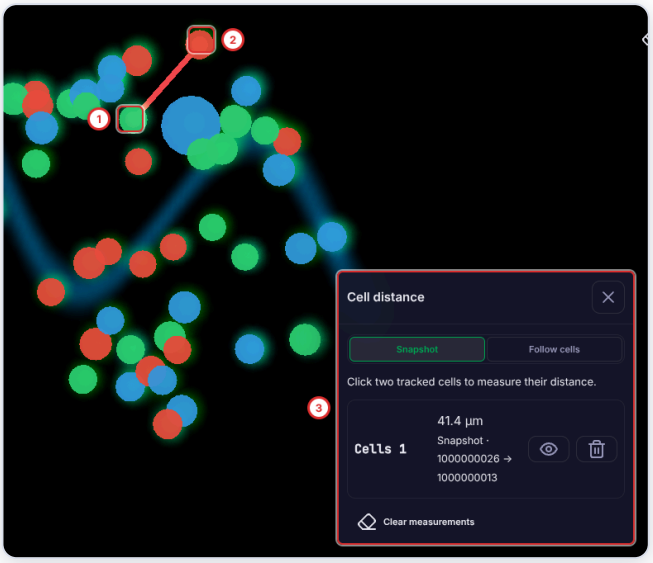
- The **timeline** plays the series at 0.5 to 20 frames per second; a bar shows the time points already loaded.
- The **cell tracking** is overlaid on the volume: cells, trajectories, embryo surface.
- Five analysis tools: trajectories, surface, cell inspector, charts, distance between cells.



▲ A tracked timelapse (demonstration dataset, 4 time points): the tracking layer and the timeline.



▲ The inspector: metrics, lineage, neighbours of a selected cell.



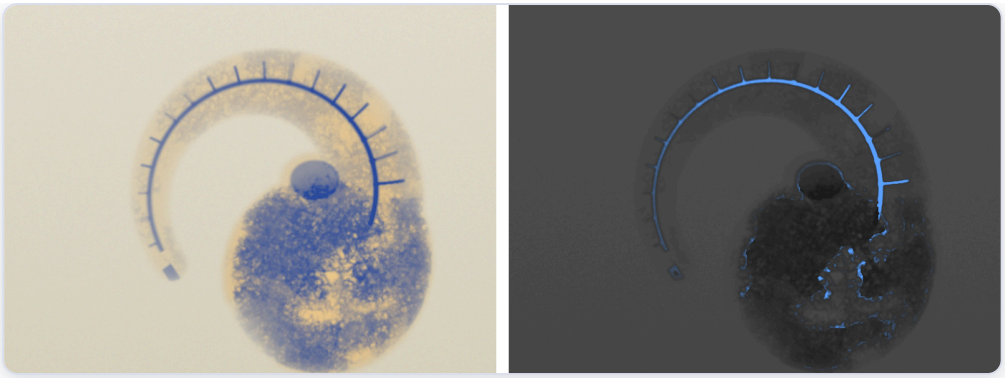
▲ The distance between two cells, followed over time.

- Playback **waits** for frames that have not arrived yet: it never silently skips a time point.
- The positions shown are the **stabilised** positions (§ 7); mitoses and fusions are marked.
- Charts (population, speed, neighbours, mitoses) and tables can be exported. → ch. 12 and 13

14. 2D photographs

⏱ **IN 30 SECONDS**

- One calibrated photograph per dataset: zoom, scale bar, measurements in µm or mm.
- **Isolate the staining** brings out the X-gal staining (blue) on the tissue (yellow).
- A contact sheet browses every photograph of the collection.



▲ Before / after “Isolate the staining”: the blue stands out, the rest turns grey (demonstration photograph).

“Isolate the staining”: three steps for every pixel

1 · Where is the yellow tissue?

a pixel's “yellowness” =
(red + green) / 2 − blue

It is averaged over a square about 40 px wide (image reduced 4 times, blur radius 5).

matte background: 2 to 6
broad yellow trunk: 6 to 9

→

2 · Is the pixel blue?

ratio blue / (red + 1)

Unstained tissue: ≈ 0.6
Strongly stained core: 2 to 4

strength v: 0 if ratio ≤ 1.0
1 if ratio ≥ 1.8, linear between

and only if the local “yellowness” exceeds 5 (otherwise v = 0)

→

3 · Repaint

grey = brightness × 0.35 × (1 − v)

red = grey + 90 × v
green = grey + 160 × v
blue = grey + 255 × v

v = 0: dark grey
v = 1: vivid blue (90, 160, 255)

Worked example
Pixel (R, G, B) = (60, 90, 170): 170 / 61 = 2.8 → v = 1, in tissue with “yellowness” 8: it turns vivid blue.
Pixel (200, 190, 120): 120 / 201 = 0.6 → v = 0: it turns dark grey (0.35 × its brightness).

▲ The computation, pixel by pixel: blue / red ratio, limited to the neighbourhood of the yellow tissue to ignore background dust.

ℹ️ **GOOD TO KNOW**

The photograph is stored as **lossy WebP (quality 90)** by default; the `--lossless` option keeps every pixel exact, useful if you want to quantify the blue / red ratio. The calibration comes from the TIFF metadata (ImageJ / Leica); without it, no scale bar. → ch. 13

15. Interpreting what you see correctly

📌 **REMEMBER**

- **Grey levels are relative.** Each channel of each dataset has its own window (floor → ceiling): an intensity cannot be compared between two channels or between two datasets. In a timelapse, the window is **shared by all time points**: a drop in intensity stays visible, as in reality.
- **Inside the signal, nothing is changed**; **outside the signal**, the noise is smoothed (3 × 3 × 3 median).
- **An object thinner than about 3 voxels** in one direction is erased by the cleaning, even a bright one (§ 4.4).
- **8 bits** are enough for morphology; to **quantify**, go back to the original `.ims` (in `download/` if the option was enabled).
- **Bricks that are not stored** are areas that really are 0 after cleaning, not lost data.
- **The display floor** (6 to 48) hides the faintest values on screen: lowering *min* will not bring them back.
- **At most 4 channels** are displayed at the same time.
- **What you see is what you measure**: on format 6 the screen shows WebP Q30 (mean error 1.5 grey levels out of 255); distances are picked on the picture on screen, so turn *Fast loading* off to measure on the lossless copy. Exports of format 6 always read the lossless copy. On format 5, the chosen column changes the picture and the distances; exports offer the lossless choice.
- **Without calibration** in the file, no measurement in µm is offered.
- **The 3D view is in perspective**: measure with the measuring tool or on a slice, not with the 3D scale bar.

15.1 What the platform does not do

NOT AVAILABLE	WHY / WHAT TO DO
Absolute intensity quantification	windowed 8-bit data: use the original `.ims`
Segmentation, automatic counting	do the analysis in Imaris or Fiji
Deconvolution	to be done before the Imaris export
Photobleaching compensation	measured by the pipeline, but not yet applied by the viewer
Editing voxels	everything is read-only; only the display can be adjusted

→ ch. 20 (appendix “What Lumen3D does not do”)

16. What happens if… ?

SITUATION	WHAT THE PLATFORM DOES	WHAT YOU CAN DO
The image is blurry at first	the coarse level arrives first, the detail follows	wait for the progress to finish
“Native” shows another level	not enough graphics memory: lighter level + message	choose 1024, close other 3D tabs
The graphics card “crashes”	the viewer reloads the view with reduced memory (at most 3 times in 2 minutes)	lower the quality, reload the page
The 3D view is refused (“no WebGL”)	the browser blocks 3D for the site after a driver crash	close the browser fully (on Android: force-stop Chrome), reopen the page
A brick is corrupt	it is shown empty and counted, never shown wrong	tell the administrator
The import transfer stops	it resumes as soon as the network is back	drag the same folder again if needed
The dataset is not in the explorer	it is hidden, or still being imported	ask the administrator to publish / show it
A measurement click is refused	nothing visible under the cursor	click on a visible structure
The Gaussian blur is greyed out	the chosen quality is too large for a single texture	choose a lower quality

→ ch. 19 (more than 60 situations in detail)

17. Words to know

Voxel

> The “pixel” of a volume: a small box, here 1.2 × 1.2 × 3.0 µm. → ch. 4

Channel

> The image of one fluorophore (DAPI, Pecam1…). → ch. 4

Floor / ceiling

> The two values that define the conversion to 8 bits: below the floor → 0, above the ceiling → 255. → ch. 5

Mask

> The voxels recognised as signal; only those outside the mask are smoothed. → ch. 5

Pyramid (levels)

> The smaller and smaller versions of the volume; level 0 is full resolution. → ch. 6

Brick

> A cube of 64³ voxels, the unit of cutting and downloading. → ch. 7

Lossless WebP

> The image format that stores the bricks while keeping every value exact. → ch. 6

WebP Q30

> The lossy copy of format 6 that the screen shows (quality 30). Format 6 exports never read it; distances are picked on the picture on screen, which uses it by default. → ch. 6

Fast loading

> The switch that shows the Q30 copies (on) or the lossless copies (off) of a format 6 dataset. → ch. 10

MIP

> Maximum intensity projection: the strongest value along an axis is kept. → ch. 7 and 9

Ray casting

> The GPU method that draws the 3D view. For each screen pixel, a ray is walked through the volume in regular steps (about one sample per voxel); each sample is read from the 3D texture and accumulated front to back, until the ray leaves the volume or, in the default Fluorescence mode, is almost opaque. → ch. 9

Gamma

> The exponent that brightens (< 1) or darkens (> 1) the mid-tones. → ch. 11

Histogram

> The chart that counts how many voxels have each value. → ch. 11

Stabilisation

> The rigid displacement that compensates for the embryo's drift in a timelapse. → ch. 8 and 13

Studio

> The workshop for annotated figures at full resolution. → ch. 12 and 13

Data format

> The version number of how a dataset's files are organised (1 to 6). → ch. 17

Matrix (quality)

> In format 5, the "resolution × compression" table of the quality panel: each cell is a way of displaying the volume. → ch. 10

📖 **GOING FURTHER**

Full documentation ("Lumen3D, from A to Z"): 21 chapters, more than 350 pages, every step illustrated. **Administrator guide**: every screen of the panel, step by step. Both are published in the *Documentation* tab of the administration panel.